

# Cognitive Aid Game for Attention and Working Memory in Children with Attention Deficit Difficulties

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GitHub Repository:

CognitiveAid GitHub Repository

Video Demo:

Primary Project Video Demonstration

Alternative Link (if primary link does not work):

Google Drive Video Demonstration

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# Abstract

This report presents a cognitive aid game designed to test and support attention and working memory in children with attention deficit difficulties. The application is implemented in Python and provides multiple gameplay modes that vary memory load, response order, spatial transformation, and symbol association. The system records detailed behavioral measures including accuracy, response time, lapse count, lapse duration, level progression, lifetime statistics, and a progress index that captures efficiency over time.

The purpose of this document is to describe the theoretical basis of the game, the aims of the project, the full set of modes and variants, the design decisions behind the interface and gameplay structure, and the value of the recorded metrics for interpretation and research. Since the system is intended for children, the design emphasizes simplicity, large targets, low distraction, and short-feedback cycles. Screenshots of the interface are intentionally left as placeholders so that the final document can be completed with actual game images from the project.

# Introduction

Children with attention deficit difficulties often experience challenges in sustained attention, selective attention, working memory, inhibitory control, and cognitive flexibility. These difficulties can affect classroom performance, task completion, and the ability to follow multi-step instructions. A game-based system is useful in this context because it can turn repeated memory and attention tasks into an engaging activity rather than a purely formal test.

The project described here is not just a game in the entertainment sense. It is also a structured behavioral task. The child is required to observe a sequence, maintain it in working memory, possibly transform it mentally, and then reproduce it under time pressure. That combination makes the task useful for observing how well the child can encode information, keep it active, resist distraction, and respond accurately.

The system is especially valuable because it can record both immediate results and accumulated session data. A single score alone does not reveal enough. In contrast, metrics such as response latency, lapse duration, level reached, and progress index allow a much richer picture of performance.

# Theoretical Background

## 3.1 Attention

Attention is the ability to focus cognitive resources on relevant information while ignoring competing distractions. In practical terms, attention determines whether a child can stay engaged with the task long enough to perceive the stimulus and hold task goals in mind. For children with attention deficit difficulties, the problem is often not a total inability to respond, but inconsistency: performance may fluctuate from one moment to the next.

The game therefore includes repeated trials and timing-based measurement. If a child performs well early in a session but later shows longer pauses, more mistakes, or slower reactions, those changes may reflect waning attention rather than lack of understanding.

## 3.2 Working Memory

Working memory is the short-term cognitive system used to temporarily store and manipulate information. Unlike passive short-term storage, working memory is active. It keeps information available while the person performs a mental operation on it. For example, repeating a sequence in reverse order or rotating a grid mentally are both working-memory tasks.

This project targets working memory in several ways:

- by requiring sequence recall,
- by increasing sequence length gradually,
- by introducing reversal,
- by introducing spatial rotation,
- by introducing symbol association,
- and by mixing multiple rules in some modes.

These variations let the same core task probe different layers of memory and executive

control.

### 3.3 Executive Function

Executive function refers to the set of higher-level processes that regulate behavior, including planning, shifting, updating, inhibition, and monitoring. In this game, executive function is needed whenever the child must do more than simply repeat the stimulus exactly as shown. Mental reversal, rotation, and mixed mode all require the child to transform the remembered information before responding.

### 3.4 Why Game-Based Assessment Helps

Traditional cognitive tests can feel repetitive or intimidating, especially to children. A game environment can reduce resistance and improve engagement. When the child is more willing to continue, the system can collect more meaningful observations.

A game also provides immediate feedback. That feedback is important because it helps the child stay aware of success and error, which can support learning and motivation. The reward structure can also improve persistence across multiple trials.

## Project Aims

The project has four main aims.

1. To create a child-friendly cognitive aid game that tests attention and working memory.
2. To implement multiple modes that assess different cognitive operations such as recall, reversal, rotation, and symbol mapping.
3. To record detailed session metrics that can be used for interpretation, comparison, and longitudinal analysis.
4. To keep the experience visually simple so that the interface itself does not create unnecessary distraction.

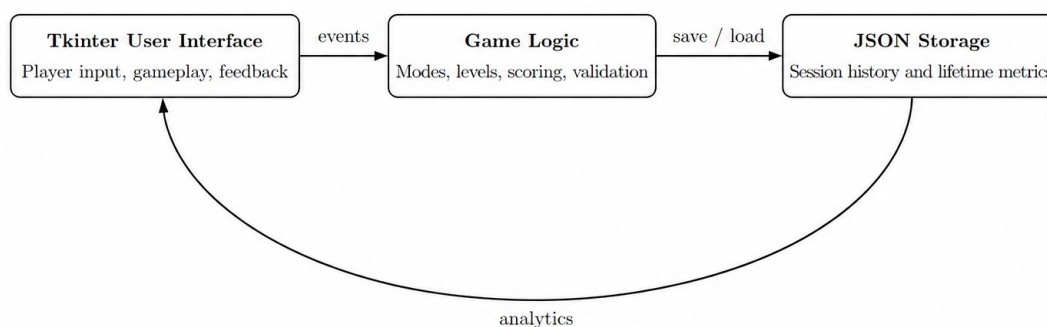
The project also has a practical aim: to build a system that can later be turned into an

app or website, allowing easier deployment and testing.

## System Overview

The application is an event-driven Python program built with Tkinter for the graphical interface. It uses local JSON storage for player and session data. The gameplay flow is organized around a start screen, a sequence presentation phase, an input phase, and a results phase.

### 5.1 High-Level Architecture



**Figure 5.1:** System architecture of the cognitive aid game.

### 5.2 Demo User Interface

The following screenshots show the implemented interface and gameplay environments of the cognitive aid system. These screenshots demonstrate the actual deployment of the project and help illustrate how the interface design supports attention and working memory assessment.

#### 5.2.1 Main Interface and Mode Selection Screen

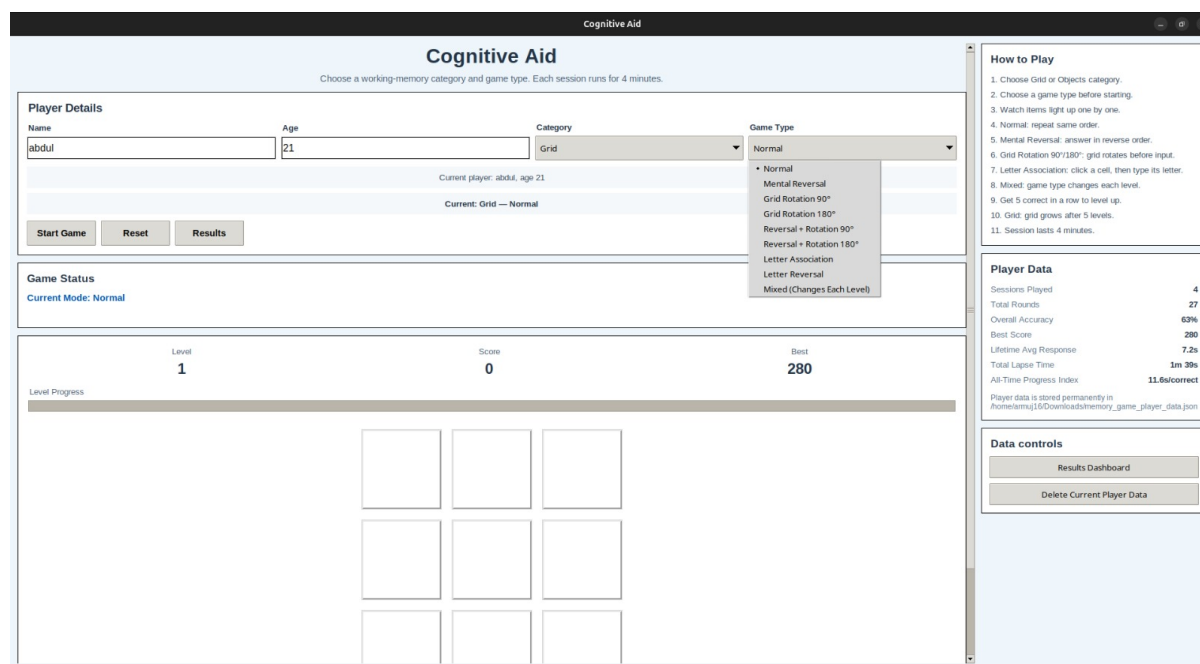
Figure 5.2 shows the main interface of the application. This screen acts as the primary interaction hub before gameplay begins. The user enters player information, selects the category type, and chooses the desired game mode.

The design intentionally uses a low-distraction visual structure. Large spacing, muted colors, and clearly separated panels reduce unnecessary cognitive clutter. This is important because the target users are children with attention deficit difficulties, who may be more

sensitive to distracting layouts.

The interface also displays cumulative player metrics such as total sessions played, overall accuracy, best score, total lapse time, and progress index. These metrics provide immediate longitudinal feedback while still keeping the gameplay region visually dominant.

The “How to Play” panel gives concise instructions for each mode. This allows the player to quickly understand the rule changes associated with reversal, rotation, and letter-association tasks.



**Figure 5.2:** Main interface of the Cognitive Aid application showing player setup, game mode selection, cumulative statistics, and the gameplay grid.

## 5.2.2 Normal Gameplay Screen

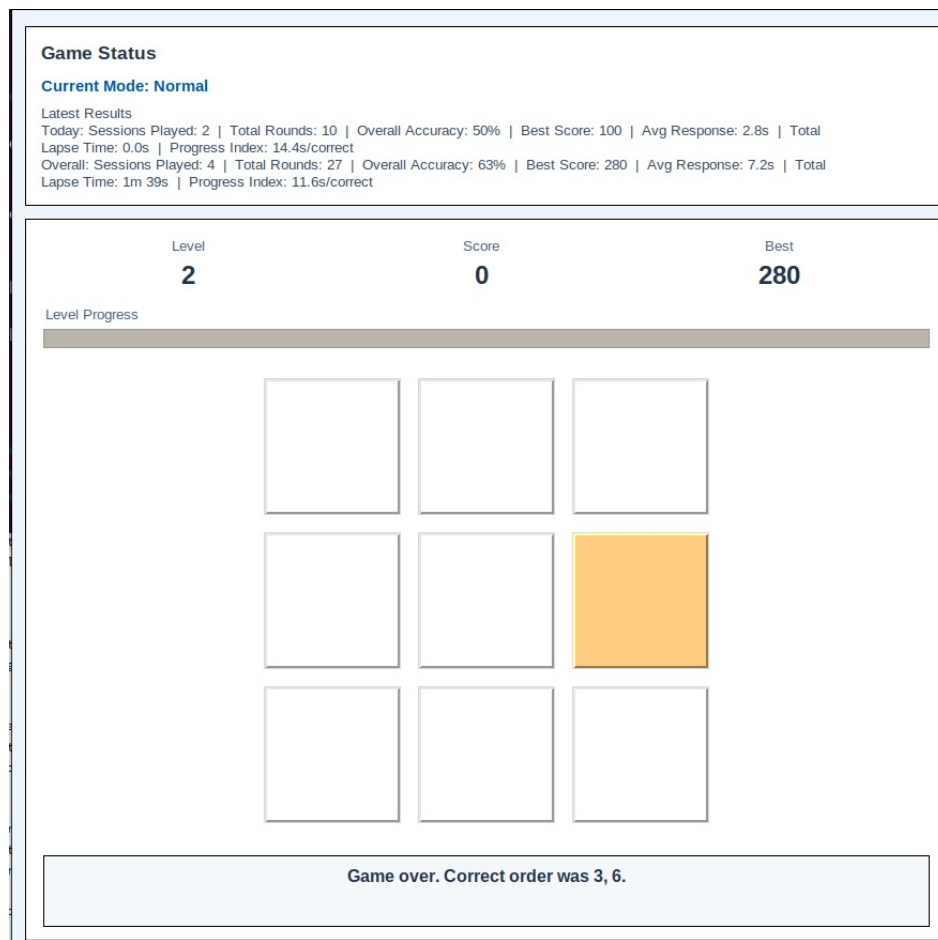
Figure 5.3 shows the standard grid-based gameplay interface during an active session. The highlighted square represents the currently active sequence item.

The layout is intentionally minimal. The central region contains only the task-relevant grid, while all supporting information is moved toward the edges of the interface. This minimizes distraction and helps sustain visual attention toward the active memory stimulus.

The level, score, and best-score indicators are positioned at the top because these are the most immediately relevant progression variables during gameplay. The progress bar provides a visual indicator of advancement through the current level hierarchy.

The “Game over” feedback message shown at the bottom is important because it provides immediate corrective information after an incorrect response. Immediate feedback helps maintain engagement and improves understanding of task expectations.





**Figure 5.3:** Normal grid-based gameplay screen during an active memory sequence trial.

### 5.2.3 Objects Category Gameplay

Figure 5.4 shows the objects category mode. Instead of standard numbered grid cells, this mode uses abstract blob-like visual objects.

The purpose of this design is to reduce monotony and introduce visually distinct stimuli that require shape-based attention rather than simple symbolic recall. The irregular shapes also increase perceptual uniqueness, which may improve memorability while still preserving task difficulty.

The floating object arrangement creates a more game-like environment and reduces the rigid appearance of traditional cognitive tasks. Despite the more playful presentation, the underlying cognitive demands remain the same: the player must encode, maintain, and reproduce the correct sequence.

The highlighted object shown in orange indicates the currently active target during sequence presentation.



**Figure 5.4:** Objects category gameplay using abstract visual stimuli instead of standard numbered grid cells.

### 5.2.4 Professional Results Dashboard

Figure 5.5 shows the professional analytics dashboard generated by the application. This dashboard transforms the game from a simple entertainment system into a meaningful behavioral analysis tool.

Several analytical visualizations are shown simultaneously:

- **Response Time Trend:** Displays how long the player took to respond across cumulative rounds. Large spikes may indicate attentional lapses, hesitation, or increasing memory load.
- **Level Progression:** Shows how the player advanced through difficulty levels over time. Sudden drops may indicate overload, fatigue, or difficulty adapting to harder tasks.
- **Accuracy by Game Mode:** Compares performance across different cognitive conditions such as Normal and Reversal modes. This helps reveal whether mental manipulation tasks are significantly harder than straightforward recall.
- **Peak Level Reached per Session:** Displays the maximum level reached in each session, providing a longitudinal view of capacity and consistency.

- **Attention Lapses:** Visualizes the duration of inactivity periods. This is especially important because lapse behavior is one of the strongest indicators of sustained attention difficulty.
- **Average Response Time by Level:** Shows how processing latency changes as the task becomes harder. Increasing latency at higher levels is expected because memory load and executive demands become larger.

The top summary row aggregates important lifetime statistics including:

- sessions played,
- total rounds,
- overall accuracy,
- best score,
- lifetime average response time,
- total lapse time,
- and progress index.

The dashboard is intentionally separated from the main gameplay environment. This ensures that the analytical interface does not distract the player during active task performance.



**Figure 5.5:** Professional results dashboard showing behavioral metrics, level progression, lapse analysis, and longitudinal statistics.

# Game Modes and Variants

The project contains several modes. Each mode is designed to stress a slightly different cognitive process.

## 6.1 Normal Mode

In Normal mode, the child watches a sequence and then reproduces it in the same order. This is the baseline condition. It mainly tests visual attention, short-term sequence retention, and response accuracy.

## 6.2 Mental Reversal Mode

In this mode, the child must repeat the sequence in reverse order. This is more demanding than Normal mode because the child cannot simply copy the sequence. The sequence must be held in memory and mentally manipulated before output.

## 6.3 Grid Rotation 90° and 180°

These modes introduce a spatial transformation. The child sees a grid and must respond as if it has been rotated. This targets visuospatial working memory and mental rotation ability. The 180-degree version is usually more demanding than the 90-degree version because the transformation is less intuitive and requires stronger internal remapping.

## 6.4 Reversal + Rotation

This is one of the most demanding conditions. The child must both reverse the order and mentally transform the grid. This mode is useful for observing whether the child can manage multiple operations simultaneously.

## 6.5 Letter Association

In this mode, each cell is associated with a letter. The child must click the correct cell and then type the corresponding letter. This tests association binding, symbol recall, and cross-modal memory.

## **6.6 Letter Reversal**

This is a harder version of letter association. The child must reproduce the letters in reverse order. This adds sequence manipulation to the symbolic association task.

## **6.7 Mixed Mode**

Mixed mode changes the active game rule across levels. This prevents the child from relying on one repetitive strategy and instead forces task switching. It is useful for observing cognitive flexibility and adaptability.

## **6.8 Objects Category**

The objects category uses abstract shapes rather than plain grid cells. The aim is to introduce a more visually rich stimulus set while still keeping the task simple. This can help reduce monotony and may reveal whether the child performs differently when the stimulus is object-like rather than number-like.

# **Detailed Design Choices and Their Effects**

## **7.1 Simple Visual Layout**

The interface uses a light background, large buttons, and limited decorative elements. This is intentional. Children with attention deficits may be more easily distracted by busy screens, so the interface avoids unnecessary clutter. The design keeps attention on the memory task rather than on the UI.

## **7.2 Large Grid Buttons**

The response targets are large and easy to click. This reduces the influence of motor precision and focuses the task on cognition. If the buttons were too small, poor performance could reflect clicking difficulty instead of memory difficulty.

## 7.3 Immediate Visual Feedback

The system gives immediate visual feedback for actions. This is important for engagement and error awareness. It helps the child understand whether the last action was correct without waiting until the end of the session.

## 7.4 Adaptive Difficulty

The game uses gradual difficulty progression. Early levels are simple, with short sequences and longer display durations. Later levels become more demanding by increasing the number of items and shortening the display and gap times. This gradual escalation helps the system remain accessible at the start while still creating sufficient challenge later.

**Table 7.1:** Difficulty progression used in the game.

| Level | Count | Show Time (ms) | Gap Time (ms) |
|-------|-------|----------------|---------------|
| 1     | 1     | 1000           | 500           |
| 2     | 2     | 920            | 430           |
| 3     | 3     | 850            | 380           |
| 4     | 4     | 780            | 330           |
| 5     | 5     | 710            | 290           |
| 6     | 6     | 650            | 250           |
| 7     | 7     | 590            | 220           |
| 8     | 8     | 530            | 200           |
| 9     | 9     | 470            | 180           |

## 7.5 Consecutive-Correct Progression

The game advances after repeated success rather than after a single correct round. This is important because one good answer could be accidental. Consecutive-correct progression gives a better picture of stable performance.

## 7.6 Recovery Behavior

When the child makes mistakes, the game can recover rather than stopping instantly. This matters because a child's performance may fluctuate due to attention, fatigue, or frustration. Recovery-based design keeps the child engaged while still preserving the

meaning of errors.

## **7.7 Limited Session Duration**

The session length is fixed to four minutes. A fixed duration has two benefits. First, it prevents the child from becoming over-fatigued. Second, it creates comparable session boundaries for analysis. This makes the recorded results easier to compare across participants or across days.

## **7.8 Randomization**

The game uses random stimulus order. This reduces predictability and ensures that the task measures real attention and memory rather than memorized patterns. Randomization also improves fairness across sessions.

## **7.9 Local Storage**

The application stores data locally in JSON format. This is simple to understand, easy to debug, and practical for offline environments. It also makes the project easier to reproduce in a laboratory or classroom setting.

## **7.10 Password-Protected Results**

The results dashboard is separated from gameplay and is protected by a password. This avoids accidental access during the game and helps preserve the clean flow of the child-facing interface.

# **Gameplay Flow**

A session proceeds in the following general order:

1. The user enters name and age.
2. The user selects category and game mode.
3. The user starts the game.
4. A sequence is shown item by item.
5. The child responds by clicking cells or typing letters.

6. The response is checked.
7. Scores, levels, progress, and other statistics are updated.
8. Session data is saved for later analysis.

This cycle repeats across rounds within the session.

## Result Metrics

This section explains the recorded metrics in detail and describes what each one reveals about attention and working memory.

### 9.1 Accuracy

Accuracy is the proportion of correct trials out of total trials.

$$\text{Accuracy} = \frac{C}{T} \times 100 \quad (9.1)$$

where  $C$  is the number of correct rounds and  $T$  is the total number of attempted rounds.

Accuracy is the most direct measure of performance. High accuracy indicates that the child can encode the sequence, maintain it, and reproduce it correctly. Low accuracy can indicate difficulty with attention, memory load, confusion about the mode, or loss of concentration.

### 9.2 Error Rate

Error rate is the complement of accuracy.

$$\text{Error Rate} = 100 - \text{Accuracy} \quad (9.2)$$

This metric is useful because it highlights failure directly. If a specific mode causes a large rise in error rate, that mode is likely imposing a stronger cognitive load.

### 9.3 Response Time

Response time is the time taken to complete a trial after the response phase begins. In this game, it is better interpreted as response latency or completion latency rather than



pure reaction time, because the child is often recalling a sequence rather than responding to a single stimulus.

$$\overline{RT} = \frac{1}{N} \sum_{i=1}^N RT_i \quad (9.3)$$

where  $RT_i$  is the response time for trial  $i$ .

A lower response time often means faster cognitive processing, but it must be interpreted carefully. Very fast responses with poor accuracy may simply reflect guessing. A good result is usually a combination of high accuracy and reasonably low response time.

## 9.4 Lapse Count

A lapse is a period of prolonged inactivity. In the current system, lapses are recorded when the session becomes inactive beyond the threshold. The lapse count is the number of such events.

$$L = \text{number of lapse events} \quad (9.4)$$

This is a strong indicator of sustained attention. A child who repeatedly stops interacting with the game may be distracted, tired, frustrated, or disengaged.

## 9.5 Total Lapse Duration

Total lapse duration is the sum of all lapse intervals.

$$T_{\text{lapse}} = \sum_{j=1}^L d_j \quad (9.5)$$

where  $d_j$  is the duration of the  $j$ th lapse.

This metric is more informative than lapse count alone because it shows not just how often attention drifted, but how long attention drifted for. Two children may have the same number of lapses, but one may have much longer pauses.

## 9.6 Longest Lapse

The longest lapse is the maximum single pause recorded in a session.

$$d_{\text{max}} = \max(d_1, d_2, \dots, d_L) \quad (9.6)$$

A long single lapse may indicate fatigue, frustration, loss of interest, or a moment of deep distraction.

## 9.7 Score

Score is the reward metric used by the game. It reflects success across rounds and contributes to the gamified experience. The exact score formula may vary depending on the implementation of reward logic, but the important point is that score gives the child immediate reinforcement.

Score is useful, but it should not be treated as the only outcome. It combines difficulty and correctness, so it is best interpreted alongside accuracy, response time, and level reached.

## 9.8 Highest Level Reached

This metric records the hardest level successfully reached during the session.

$$L_{\max} = \max(\text{levels reached in session}) \quad (9.7)$$

This is a useful estimate of the child's current memory capacity under the game rules. A higher value suggests the child can maintain performance at greater difficulty.

## 9.9 Progress Bar Versus Progress Index

The interface includes a progress bar that shows advancement through the current difficulty ladder. This is a visual UI element.

This should not be confused with the progress index. The progress bar is a display of level advancement. The progress index is a calculated performance metric that reflects efficiency.

## 9.10 Progress Index

The progress index is one of the most useful summary metrics in the system. It is designed to combine correctness, attention stability, and time efficiency into a single measure.

$$\text{Progress Index} = \frac{T_{\text{active}}}{C} \quad (9.8)$$

where

$$T_{\text{active}} = T_{\text{session}} - T_{\text{lapse}} \quad (9.9)$$

and  $C$  is the number of correct responses.

In the code, this value is displayed in seconds per correct response. A lower progress index is better because it means the child is achieving correct responses with less active time. A higher progress index suggests slower processing, more hesitation, or less efficient sustained performance.

This metric is especially valuable because it captures more than score alone. For example, two children might have the same accuracy, but one may complete correct trials more quickly and with fewer lapses. The progress index would distinguish between them.

## 9.11 Lifetime Metrics

The application also stores lifetime metrics across sessions. These include:

- total sessions played,
- total rounds played,
- total rounds correct,
- overall accuracy,
- best score,
- average response time,
- total lapse count,
- total lapse duration,
- average lapse duration,
- longest lapse,
- lifetime progress index.

These metrics are important for longitudinal analysis. They allow the researcher to see whether the child improves across repeated sessions, whether the child becomes faster or

more accurate, and whether attentional lapses decrease.

## How the Results Can Be Analyzed

### 10.1 Performance Across Modes

One useful analysis is to compare performance across the different modes. For example, if Normal mode has high accuracy but Reverse mode has much lower accuracy, that suggests the child has more difficulty with mental manipulation than with simple recall.

A similar comparison can be made between 90-degree and 180-degree rotation modes. If 180-degree rotation is consistently harder, then the data support the idea that larger spatial transformations require more cognitive effort.

### 10.2 Difficulty Versus Accuracy

As levels increase, accuracy should generally decrease if the game is successfully creating more difficult conditions. A flat curve may mean the levels are not distinct enough, while an abrupt drop may mean the difficulty increases too sharply.

### 10.3 Difficulty Versus Response Time

Higher levels may produce longer response times because the child needs more time to hold the sequence in memory and plan the correct answer. If response time grows while accuracy remains stable, the child may be compensating successfully. If both response time and error rate rise together, the task may be too demanding.

### 10.4 Lapses as Attention Indicators

Lapses are especially useful because they reveal attention problems that may not appear as direct incorrect answers. A child may still answer some sequences correctly, but frequent lapses can show that sustaining engagement is difficult.

### 10.5 Progress Index as an Efficiency Measure

The progress index is especially helpful for comparing overall efficiency across participants or sessions. A lower progress index suggests better performance efficiency. A rising progress index across a session may suggest fatigue or disengagement. A falling progress

index across multiple sessions may indicate improvement.

## 10.6 Longitudinal Learning

If the same child plays multiple sessions, the data can show whether practice improves performance. Improvement may appear as:

- higher accuracy,
- lower response time,
- fewer lapses,
- higher level reached,
- and a lower progress index.

That would support the game's usefulness not only as a measurement tool but also as a cognitive aid.

# Implementation Notes

The game is written as a single Python application. The program uses event handlers for clicks and keyboard input, scheduled callbacks for sequence display, and local data saving for persistence. The structure is practical and straightforward for future expansion.

The code already supports several important internal features:

- multiple game modes,
- progressive difficulty,
- local player profiles,
- per-session and lifetime statistics,
- results dashboard access,
- and data persistence.

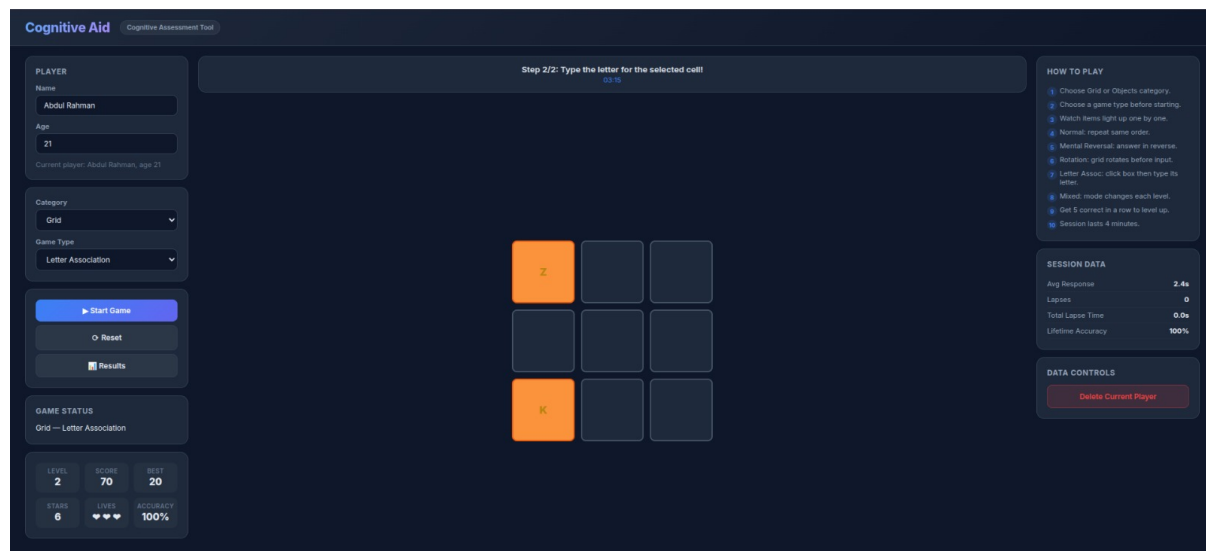
These make the project more than a simple memory game. They make it a framework for

cognitive interaction and behavioral observation.

# Future Work

## 12.1 Better UI and App or Website Deployment

A major future goal is to redesign the interface and deploy the project as a proper app or website. Preliminary work in that direction has already begun. A polished interface would make the system easier to use, better suited for children, and more appropriate for real-world testing.



**Figure 12.1:** An example image of the User Interface being developed

## 12.2 Multiple Screen Flow

The application can be improved by separating the experience into multiple screens. A possible flow is:

- login or player entry screen,
- short instructions screen,
- distraction-free gameplay screen,
- results screen at the end,
- and optional analytics screen for parents, teachers, or researchers.

This would help because children with attention deficit difficulties benefit from a clear and

focused presentation. Too much text or too many metrics on the main gameplay screen could distract them. A dedicated game screen keeps attention centered on the task.

## **12.3 Sound Cues**

Future versions should include sound cues for correct trials, incorrect trials, rotation events, selection events, and mode transitions. Sound can provide immediate reinforcement and can make the game easier to follow. It may also improve engagement by giving the child another channel of feedback besides vision.

## **12.4 Real-World Testing and Cognitive Evaluation**

A major future direction is testing the system with real users, particularly children and individuals diagnosed with ADHD or related attention difficulties. Real-world testing would help evaluate how accurately the recorded metrics such as accuracy, lapse duration, response latency, level progression, and progress index reflect sustained attention and working-memory performance across repeated sessions.

Longitudinal studies could also explore whether repeated interaction with the game helps improve attention stability, sequence recall, and executive-function skills over time. Comparing performance trends across different user groups may further help identify which gameplay modes are most sensitive to attentional and working-memory differences in practical settings.

## **12.5 Additional Future Directions**

Other valuable improvements include:

- touch-friendly mobile controls,
- browser-based deployment,
- automatic report export,
- adaptive difficulty based on current accuracy,
- therapist or parent dashboards,
- and richer stimulus types such as colors, patterns, or animated shapes.

# Limitations

This project is promising, but it also has limitations.

First, the current application is a prototype, so the metrics are more useful as behavioral indicators than as clinically validated diagnostic measures. Second, the interface still needs refinement for deployment as a polished product. Third, some scores and thresholds are heuristic and may need empirical tuning. Fourth, formal validation against standard cognitive tests is still needed before strong scientific claims can be made.

These limitations do not reduce the value of the project. Instead, they define the next stage of development.

# Personal Learnings and Insights

Developing this project provided valuable insights into the relationship between cognitive psychology and interactive system design. One of the most important realizations during development was that attention and working memory are highly sensitive to interface complexity, timing, feedback style, and task structure. Even small changes in sequence speed, visual clutter, or mode-switching significantly affected cognitive load and user performance. Designing gameplay modes such as reversal, rotation, and letter association also helped illustrate the distinction between simple short-term recall and active working-memory manipulation involving executive control and mental transformation.

The project additionally highlighted how behavioral metrics such as response latency, lapse duration, level progression, and progress index can reveal patterns that are not visible through score alone. Observing these trends reinforced the importance of multidimensional cognitive analysis, particularly when designing systems intended for children with attention deficit difficulties. From a technical perspective, the work strengthened practical understanding of event-driven GUI programming, adaptive gameplay systems, data persistence, and longitudinal performance tracking, while also demonstrating how game-based systems can be designed as meaningful cognitive-assistance tools rather than



purely entertainment applications.

# Reproducibility

The project is available through the GitHub repository listed in the title page. The system is reproducible because it is a local Python application with transparent logic and local data storage. A typical setup is:

```
# Clone the repository
git clone https://github.com/abdulrm624/CognitiveAid.git

# Move into the project directory
cd CognitiveAid

# Create a virtual environment
python3 -m venv cognitive-aid

# Activate the virtual environment
source cognitive-aid/bin/activate

# Install required libraries
pip install pillow matplotlib

# If tkinter is missing on Linux
sudo apt install python3-tk

# Run the application
python3 btp1.py
```

The local JSON storage makes it easy to inspect saved sessions and lifetime data. This is useful for debugging, reporting, and analysis.

# Conclusion

This project demonstrates how a simple game can be structured as a cognitive aid for children with attention deficit difficulties. The game is designed to test sustained attention, working memory, executive control, mental reversal, rotation, and associative recall. The

metrics collected by the system make it possible to move beyond a simple score and analyze attention lapses, response latency, difficulty progression, and overall efficiency through the progress index.

The design choices were made to keep the game child-friendly while preserving analytical value. The result is a system that is both engaging and scientifically meaningful. With future improvements in interface design, multi-screen flow, sound cues, and web/app deployment, the project can become even more effective as a cognitive assessment and training tool.